```
import pandas as pd
import matplotlib.pyplot as plt
import plotly.express as px

sales = pd.DataFrame({
    "Month": ["Jan", "Feb", "Mar", "Apr", "May", "Jun",
              "Jul", "Aug", "Sep", "Oct", "Nov", "Dec"],
    "Laptops":    [40, 42, 45, 38, 36, 35, 41, 48, 52, 85, 95, 60],
    "Phones":     [55, 54, 58, 60, 57, 59, 62, 64, 66, 90, 98, 72],
    "Headphones": [20, 22, 21, 25, 24, 26, 25, 27, 30, 45, 52, 40],
})

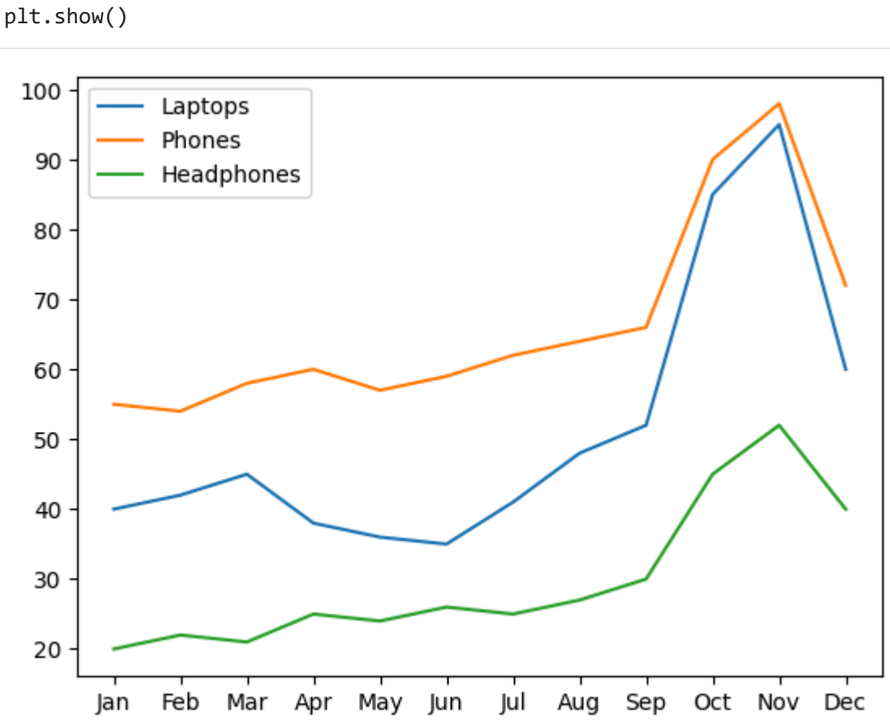
sales
```

	Month	Laptops	Phones	Headphones
0	Jan	40	55	20
1	Feb	42	54	22
2	Mar	45	58	21
3	Apr	38	60	25
4	May	36	57	24
5	Jun	35	59	26
6	Jul	41	62	25
7	Aug	48	64	27
8	Sep	52	66	30
9	Oct	85	90	45
10	Nov	95	98	52
11	Dec	60	72	40

Next steps: Generate code with `sales` New interactive sheet

Part A: The Starting Point Task 1: Default chart • Draw a line chart of Laptops, Phones and Headphones against Month with default settings and a legend. • Save it using plt.savefig("before.png"). Do not change anything yet.

```
plt.figure()
plt.plot(sales['Month'], sales['Laptops'], label='Laptops')
plt.plot(sales['Month'], sales['Phones'], label='Phones')
plt.plot(sales['Month'], sales['Headphones'], label='Headphones')
plt.legend()
plt.savefig('before.png')
plt.show()
```



Part B: Colours and Style Task 2: Consistent, readable colours • Apply a clean style with plt.style.use("seaborn-v0_8-whitegrid"). • Draw the same chart again, this time using fixed colours for each product, a line width of 2.5 and circle markers: colors = {"Laptops": "#0072B2", "Phones": "#D55E00", "Headphones": "#009E73"} • In 2 lines, explain why a product should keep the same colour in every chart, and why red and green should not be used together.

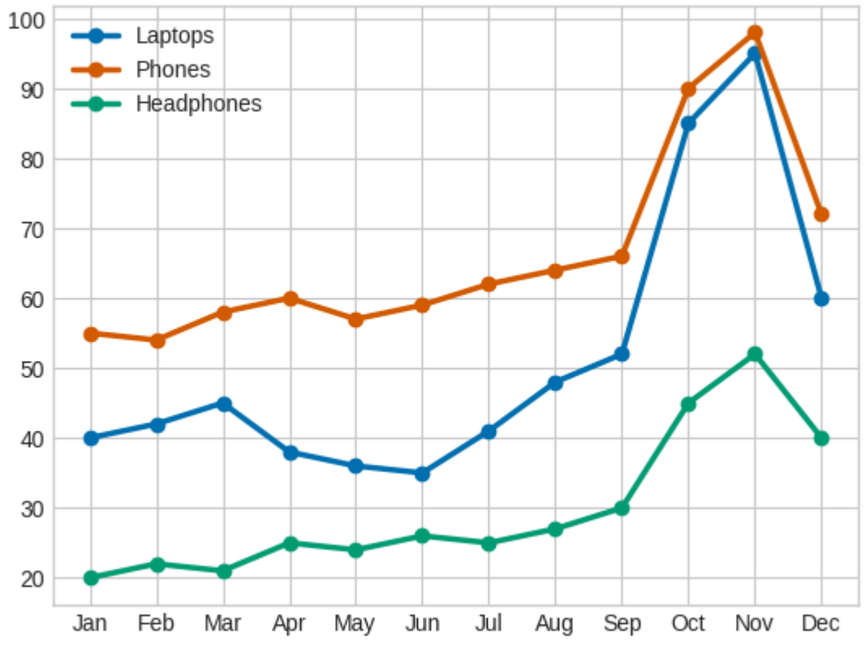
```
import matplotlib.pyplot as plt

plt.style.use("seaborn-v0_8-whitegrid")

colors = {"Laptops": "#0072B2", "Phones": "#D55E00", "Headphones": "#009E73"}

plt.figure()
plt.plot(sales['Month'], sales['Laptops'], color=colors['Laptops'], linewidth=2.5, marker='o', label
plt.plot(sales['Month'], sales['Phones'], color=colors['Phones'], linewidth=2.5, marker='o', label='
plt.plot(sales['Month'], sales['Headphones'], color=colors['Headphones'], linewidth=2.5, marker='o',
plt.legend()
plt.show()

# Explanation:
# Consistent product colors across charts prevent cognitive load and help users immediately identify
# Red and green should not be used together to ensure accessibility for colorblind individuals, who
```



Part C: Titles and Labels Task 3: Say what the chart shows • Replace the title with one that states the insight, for example "Phones lead all year, and every product peaks in November". Use ax.set_title(..., fontsize=14, fontweight="bold", loc="left"). • Label the y-axis with its unit: "Sales (Rs. thousands)". • Remove the legend. Instead, write each product's name at the end of its line using ax.text(11.2, last_value, name, color=colour). • Remove the top and right borders using ax.spines[["top", "right"]].set_visible(False).

```
import matplotlib.pyplot as plt

colors = {"Laptops": "#0072B2", "Phones": "#D55E00", "Headphones": "#009E73"}

fig, ax = plt.subplots(figsize=(8, 5))

ax.plot(sales['Month'], sales['Laptops'], color=colors['Laptops'], linewidth=2.5, marker='o')
ax.plot(sales['Month'], sales['Phones'], color=colors['Phones'], linewidth=2.5, marker='o')
ax.plot(sales['Month'], sales['Headphones'], color=colors['Headphones'], linewidth=2.5, marker='o')

ax.set_title("Phones lead all year, and every product peaks in November", fontsize=14, fontweight="bold", loc="left")

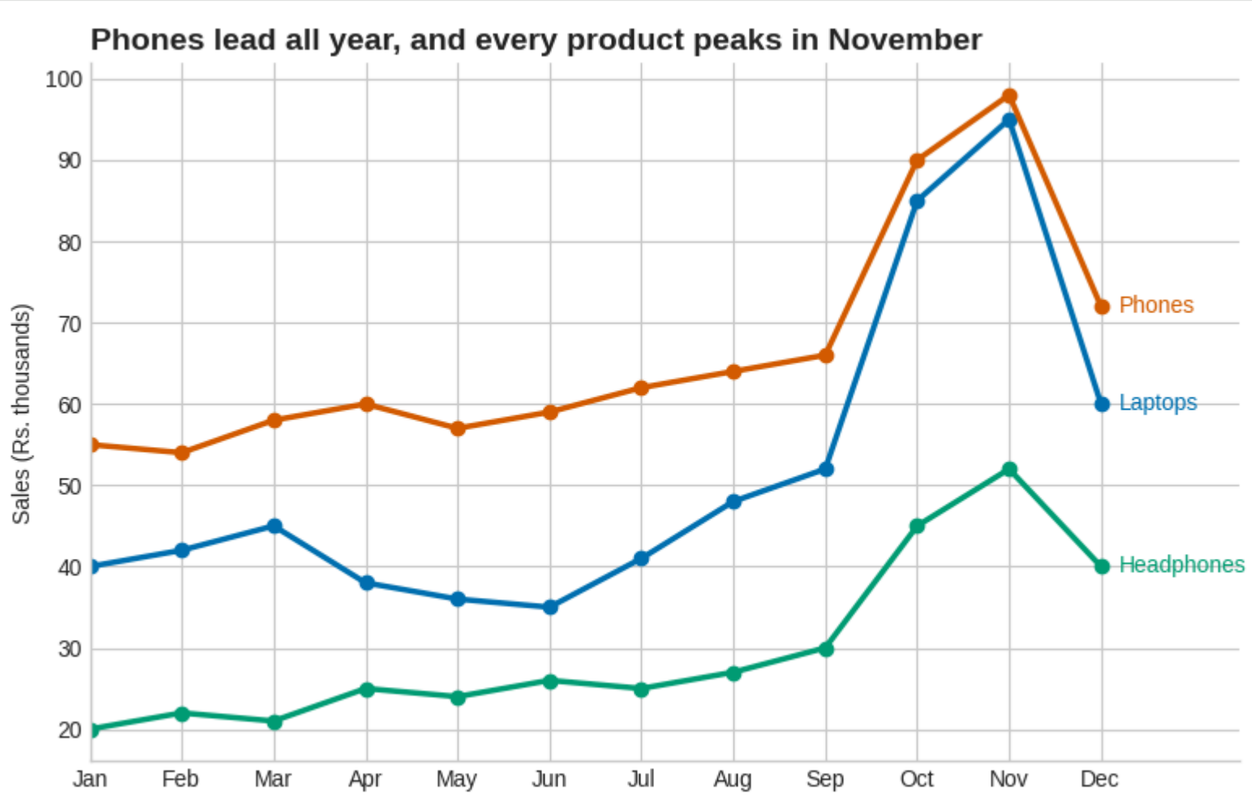
ax.set_ylabel("Sales (Rs. thousands)")

ax.text(11.2, sales['Laptops'].iloc[-1], "Laptops", color=colors['Laptops'], va='center')
ax.text(11.2, sales['Phones'].iloc[-1], "Phones", color=colors['Phones'], va='center')
ax.text(11.2, sales['Headphones'].iloc[-1], "Headphones", color=colors['Headphones'], va='center')

ax.set_xlim(0, 12.5)

ax.spines[["top", "right"]].set_visible(False)

plt.tight_layout()
plt.show()
```



Part D: Annotations and Highlights Task 4: Guide the reader's eye • Add an arrow with text pointing to the highest point using ax.annotate("Festive season peak", xy=(10, 98), xytext=(5.5, 90), arrowprops=dict(arrowstyle="->")). • Shade October and November using ax.axvspan(8.5, 10.5, color="orange", alpha=0.2). • Add a dashed grey line at 60 labelled "Monthly target" using ax.axhline(60, linestyle="--", color="grey") and ax.text(). • Save the final chart with plt.savefig("after.png", dpi=150, bbox_inches="tight").

```
import matplotlib.pyplot as plt

plt.style.use("seaborn-v0_8-whitegrid")
colors = {"Laptops": "#0072B2", "Phones": "#D55E00", "Headphones": "#009E73"}

fig, ax = plt.subplots(figsize=(8, 5))

ax.plot(sales['Month'], sales['Laptops'], color=colors['Laptops'], linewidth=2.5, marker='o')
ax.plot(sales['Month'], sales['Phones'], color=colors['Phones'], linewidth=2.5, marker='o')
ax.plot(sales['Month'], sales['Headphones'], color=colors['Headphones'], linewidth=2.5, marker='o')

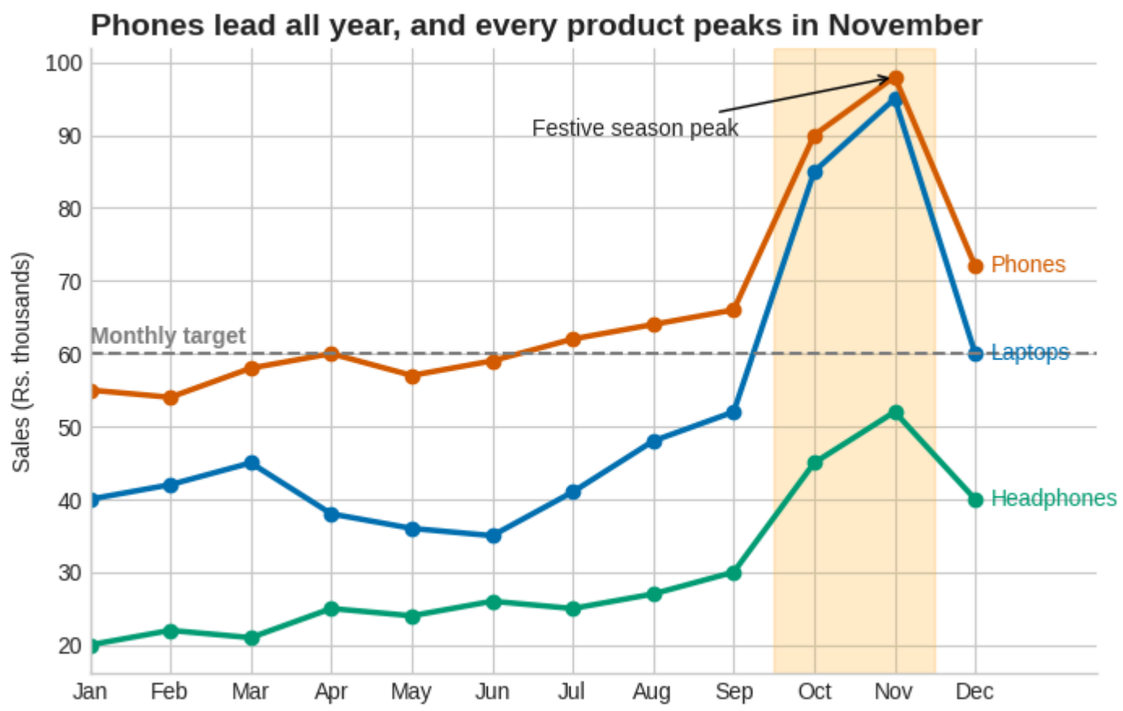
ax.set_title("Phones lead all year, and every product peaks in November", fontsize=14, fontweight="bold", loc="left")
ax.set_ylabel("Sales (Rs. thousands)")
ax.text(11.2, sales['Laptops'].iloc[-1], "Laptops", color=colors['Laptops'], va='center')
ax.text(11.2, sales['Phones'].iloc[-1], "Phones", color=colors['Phones'], va='center')
ax.text(11.2, sales['Headphones'].iloc[-1], "Headphones", color=colors['Headphones'], va='center')
ax.set_xlim(0, 12.5)
ax.spines[["top", "right"]].set_visible(False)

ax.annotate("Festive season peak", xy=(10, 98), xytext=(5.5, 90),
            arrowprops=dict(arrowstyle="->", color="black"))

ax.axvspan(8.5, 10.5, color="orange", alpha=0.2)
```

```
ax.axhline(60, linestyle="--", color="grey")
ax.text(0, 61.5, "Monthly target", color="grey", fontweight="semibold")

plt.savefig("after.png", dpi=150, bbox_inches="tight")
plt.show()
```



Task 5: Highlight one bar • Draw a bar chart of monthly Phones sales. Colour every bar light grey except the month with the highest sales, which should be crimson. Hint: find it with sales["Phones"].idxmax(). • Show the value above each bar using ax.bar_label(). • Give the chart a title that states the insight and remove the top and right borders. • In one line, explain why this makes the reader look at the right bar first.

```
import matplotlib.pyplot as plt

max_idx = sales["Phones"].idxmax()

bar_colors = ["crimson" if i == max_idx else "#D3D3D3" for i in range(len(sales))]

fig, ax = plt.subplots(figsize=(8, 5))

bars = ax.bar(sales["Month"], sales["Phones"], color=bar_colors)

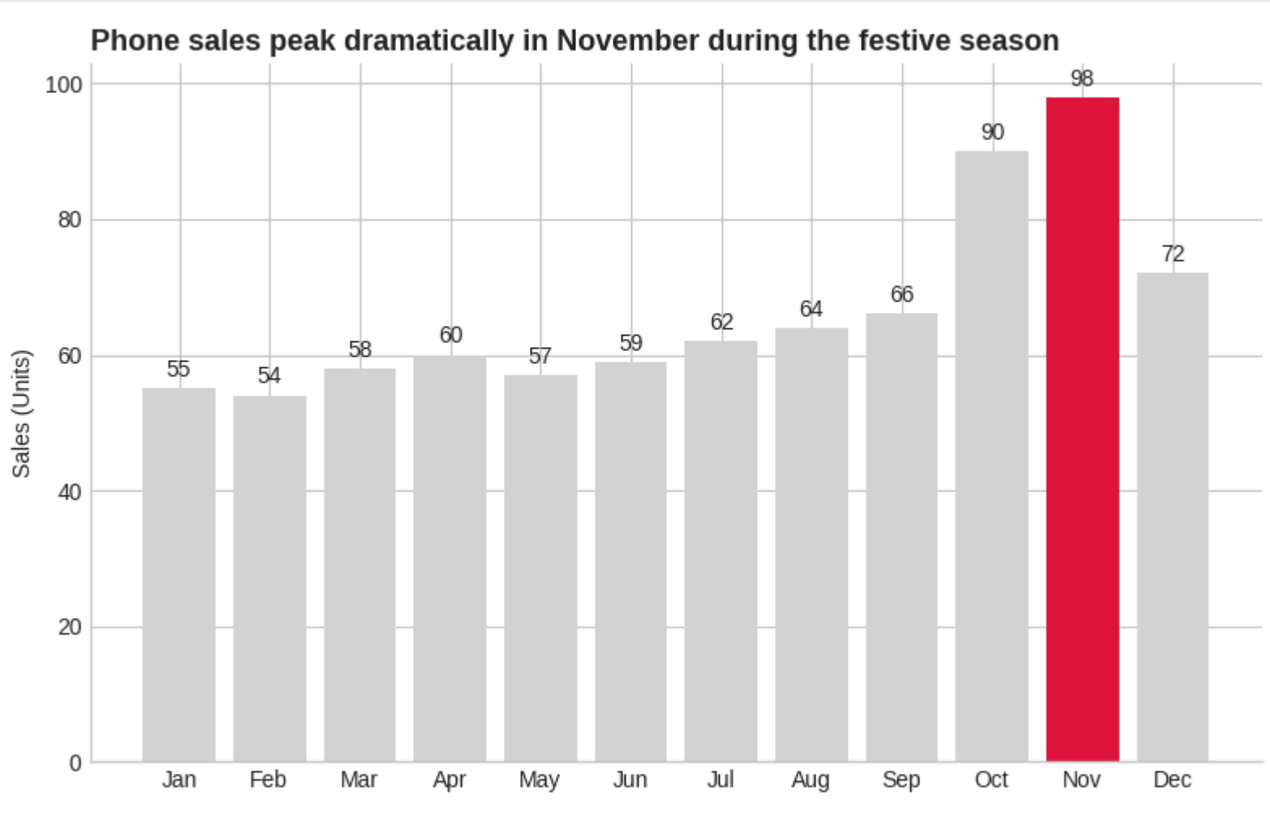
ax.bar_label(bars, padding=3)

ax.set_title("Phone sales peak dramatically in November during the festive season", fontsize=13, fontweight="bold", loc="left")
ax.set_ylabel("Sales (Units)")

ax.spines[["top", "right"]].set_visible(False)

plt.tight_layout()
plt.show()

# Explanation:
# Using a highly contrasting color like crimson against desaturated light grey creates immediate visual salience, guiding the reader's focal point directly to the peak month first
```



Part E: Customizing a Plotly Chart Task 6: Interactive version • Convert the data to long format and draw the chart: long = sales.melt(id_vars="Month", var_name="Product", value_name="Sales")

fig = px.line(long, x="Month", y="Sales", color="Product", markers=True, color_discrete_map=colors, template="simple_white", title="Phones lead all year, and every product peaks in November") fig.show() • Make the lines thicker and change the hover text with fig.update_traces(line_width=3, hovertemplate="%{x}: Rs. %{y}k"). • Change the y-axis title using fig.update_layout(yaxis_title="Sales (Rs. thousands)"). • Add an annotation with fig.add_annotation(x="Nov", y=98, text="Festive season peak", showarrow=True, arrowhead=2). • Shade October to November with fig.add_vrect(x0="Oct", x1="Nov", fillcolor="orange", opacity=0.2, line_width=0).

```
import plotly.express as px

long = sales.melt(id_vars="Month", var_name="Product", value_name="Sales")

colors = {"Laptops": "#0072B2", "Phones": "#D55E00", "Headphones": "#009E73"}

fig = px.line(long, x="Month", y="Sales", color="Product", markers=True,
              color_discrete_map=colors, template="simple_white",
              title="Phones lead all year, and every product peaks in November")

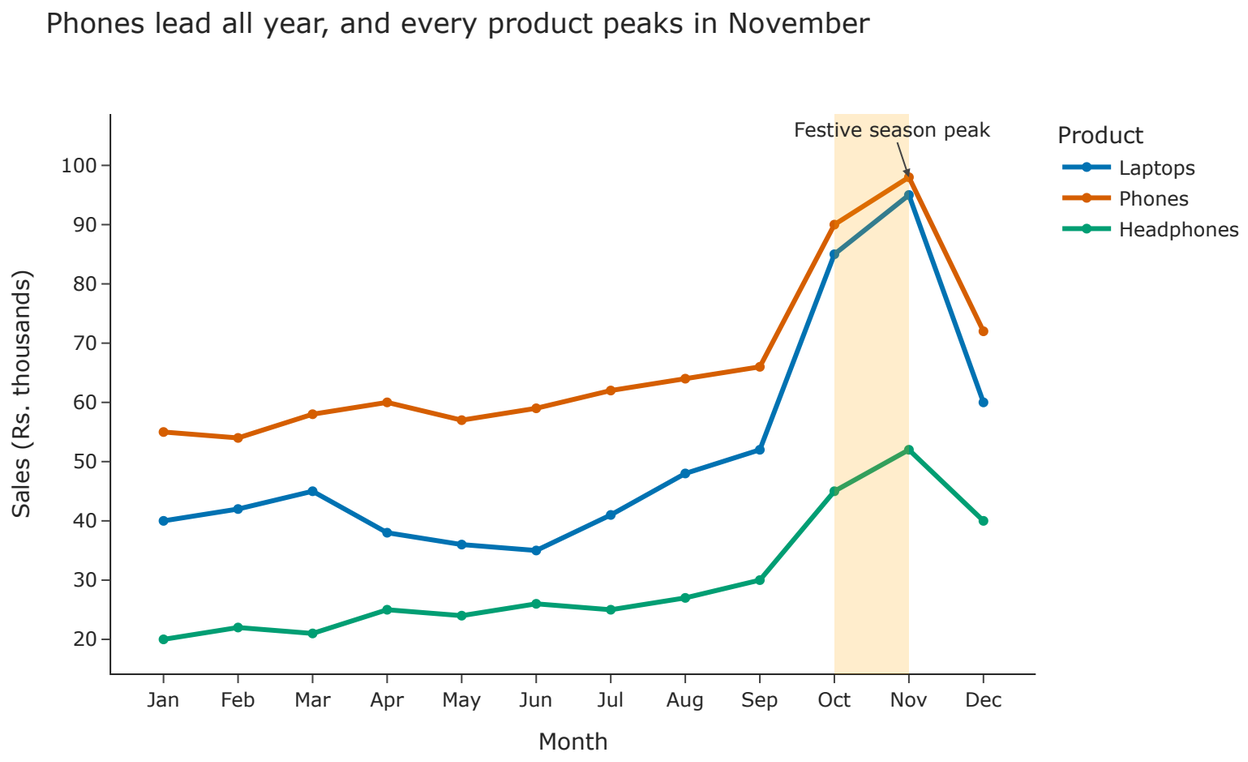
fig.update_traces(line_width=3, hovertemplate="%{x}: Rs. %{y}k")

fig.update_layout(yaxis_title="Sales (Rs. thousands)")
```

```
fig.add_annotation(x="Nov", y=98, text="Festive season peak", showarrow=True, arrowhead=2)

fig.add_vrect(x0="Oct", x1="Nov", fillcolor="orange", opacity=0.2, line_width=0)

fig.show()
```



Part F: From Chart to Story Task 7: Before and after • Show before.png and after.png side by side in your notebook. • List 5 changes you made and, for each one, one line on how it helps the reader.

```
import matplotlib.pyplot as plt
import matplotlib.image as mpimg

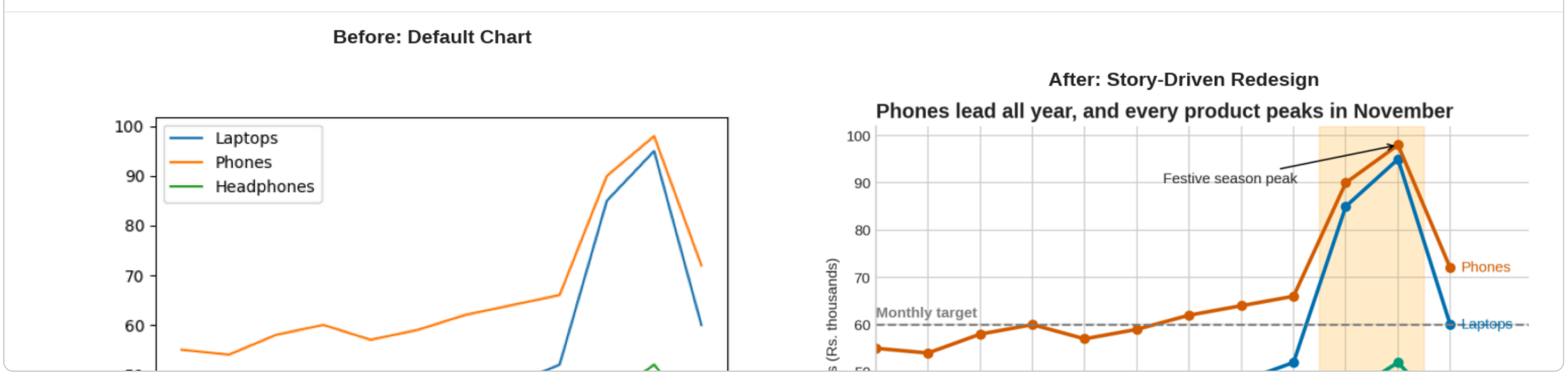
img_before = mpimg.imread('before.png')
img_after = mpimg.imread('after.png')

fig, axes = plt.subplots(1, 2, figsize=(15, 7))

axes[0].imshow(img_before)
axes[0].set_title("Before: Default Chart", fontsize=14, fontweight="bold")
axes[0].axis('off')

axes[1].imshow(img_after)
axes[1].set_title("After: Story-Driven Redesign", fontsize=14, fontweight="bold")
axes[1].axis('off')

plt.tight_layout()
plt.show()
```



5 Key Design Changes and Their Benefits

1. **Colorblind-Friendly Color Palette**
   - **How it helps:** Replacing default colors with a vetted palette ensures individuals with red-green color blindness can easily tell the product lines apart.
2. **Direct Labeling instead of a Legend**
   - **How it helps:** Placing names ("Phones", "Laptops", "Headphones") directly next to the end of each trendline eliminates the need for the reader to look back and forth between a legend and the lines.
3. **Insight-Focused, Left-Aligned Title**
   - **How it helps:** A descriptive title starting at the top-left tells the reader what insight to take away from the chart instantly, rather than forcing them to decipher the message.
4. **Shaded Region & Directed Arrow Annotation**
   - **How it helps:** Shading the October-November region and pointing an arrow to the highest peak immediately directs visual attention to the critical "Festive season peak" event.
5. **Cleaned Spines and Grid (Removing Clutter)**
   - **How it helps:** Removing the unnecessary top/right borders and adopting a softer white grid reduces visual noise and maximizes the data-ink ratio.

Task 8: Write the data story Write 5 to 6 lines for the store manager in this order: • Headline: one sentence with the main insight. • Evidence: two facts with numbers taken from your chart. • Recommendation: one action the store should take.

Data Story: Preparing for the Festive Season Surge

- **Headline:** Phones lead our electronics category throughout the entire year, culminating in a dramatic, cross-product sales peak during the November festive season.

- **Evidence 1:** Phones maintained the highest sales volume in every month, reaching an annual peak of 98,000 units sold in November.
- **Evidence 2:** This November surge was a universal trend across all product groups, with Laptops and Headphones simultaneously climbing to their highest respective yearly outputs of 95,000 and 52,000 units.
- **Recommendation:** The store should proactively scale up inventory levels and increase staffing capacity beginning in late September to fully capitalize on this high-volume October-November promotional window.